

Help build a personal AI computer.

MOB is a voice-led, battery-powered Linux computer with its own local language model and agent harness. We are seeking engineering, component, battery and manufacturing partners to turn a clear product concept into a validated prototype.

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Project: <https://mob-agent-network.vabbyshabbyy.chatgpt.site>

Location: Kota, India

Current stage

Concept and feasibility. Published component dimensions inform the target enclosure, but there is no validated integrated hardware prototype or completed mechanical stack. Battery capacity, runtime, thermal behavior and Bluetooth AI call handling remain unverified. No supplier is contracted through this request.

Requested scope

Please identify which work packages you can own: compute/BSP and model enablement; carrier electronics and power; duplex Bluetooth call audio; mechanical/thermal design; custom protected battery; DFM/DFT, assembly, programming, calibration and final test.

Planning quantities

Stage	Quantity for quotation	Purpose
Bench / EVT	10-25	Feasibility and engineering validation
DVT	100	Design, reliability and compliance validation
PVT / first production	500-1,000	Process validation; conditional on demand and readiness
Sensitivity	5,000	Volume economics comparison only

This is a non-binding RFI, not a purchase order or a statement that funding is secured. Please separate feasibility, NRE, tooling and recurring unit prices.

A real component starting point.

Subsystem	Candidate / published size	Work still required
Compute	Lantronix Open-Q 8550CS (QCS8550) 16 GB / 128 GB option 54 x 45 x 3.61 mm	Confirm BSP, model/QNN path, lifecycle, price, dev-kit availability and power
Calls	Microchip BM83 32 x 15 x 2.5 mm	Prove full-duplex PCM/I2S path and phone routing; firmware bridge and AEC
Microphones	2 x Infineon IM69D128S 3.5 x 2.65 x 1.0 mm each	PDM capture, cavity, noise, wake word and barge-in
Speaker	1 W / 8 ohm reference Adafruit 3923, 30 x 20 x 5 mm	Select production part and acoustic chamber
Battery	Custom 7,000 mAh / 3.7 V target 90 x 60 x 12 mm allocation	Capacity/size must be confirmed by cell vendor; protection, NTC and swelling
Charging	USB-C primary; Qi receiver candidate TI BQ51013C	Power-path/PD integration, coil/ferrite and thermal validation

Mechanical targets

112 x 72 x 26 mm target enclosure. White/silver finish, protected crown, physical mic cutoff, front speaker/status indicator, magnetic rear, serviceable battery, and approximately 100 mm stowed USB-C lead. The enclosure is a target, not a validated fit.

Allow for LGA/carrier height, antenna keep-outs, battery swelling, USB strain relief, audio cavity, ferrite/magnet stack, shielding, fasteners, gasket and tolerances. Explore heat spreading plus a small blower/duct if measurements require it. Keep vents clear when phone-mounted. Do not route inference heat into the cell or phone.

Prove the hard parts before freezing design.

Local compute and software

Run a small quantized language model, local speech recognition/synthesis, browser and command harness on the onboard Linux computer. Confirm the exact BSP and supported inference toolchain, memory headroom and licenses. Quote engineering support for model enablement rather than assuming QCS8550 NPU support for every model.

Calls and control

Demonstrate a real iPhone HFP call with intelligible duplex audio, echo cancellation, barge-in, call-state handling and handset takeover. Include call waiting and fault cases. The microphone hardware cutoff and stop/takeover path must remain effective when the application or model stalls.

Power and thermal budget

$7\text{ Ah} \times 3.7\text{ V} = 25.9\text{ Wh}$ nominal. With an assumed 85% usable energy, 12 hours requires average system power at or below 1.83 W. An illustrative mixed-use budget is $10\text{ h} \times 0.7\text{ W} + 2\text{ h} \times 6\text{ W} = 19\text{ Wh}$; continuous 6 W implies about 3.7 h. These are not measured values. Provide your assumptions and the test plan needed to accept or reject this target.

Device network and iOS

Wi-Fi or phone hotspot supplies internet; no cellular modem in this baseline. Tailscale connects enrolled nodes. Each remote target needs an authorized companion/SSH/API adapter and OS permissions. The iOS app supports setup, approvals and receipts; it does not grant unrestricted app access.

Validation deliverables

Area	Required evidence
Bring-up	Power rails, boot, recovery, compute/speech benchmark and thermal traces
Reliability	Battery protection, charge/load tests, mic cutoff, cancellation and call takeover
Production	DFM/DFT review, programming/calibration fixture, serial traceability and golden unit
Compliance planning	Applicable market-specific RF, electrical/battery safety and transport plan; lab scope and costs separately quoted

A comparable proposal. A concrete next step.

Please return

1. Named commercial and technical owners, relevant shipped-product examples, and the work packages you can deliver.
2. Feasibility concerns, proposed alternatives and the minimum prototype path.
3. Separate NRE, tooling, fixture, software/license and unit pricing at 25 / 100 / 1,000 / 5,000 units.
4. MOQ, sample availability, long-lead items, build lead time, quotation validity, payment milestones and shipping terms.
5. BSP/source access, engineering support, lifecycle/change-notification policy, warranty and repair support.
6. Manufacturing process, quality system, test coverage, traceability, yield assumptions and failure disposition.
7. Deliverable and IP/tooling ownership terms, assumptions, dependencies and exclusions.

Battery-specific response

Provide an actual dimensioned cell/pack drawing, nominal and minimum capacity, discharge curves, continuous/peak current, charge limits, cycle expectations, thickness tolerance and swelling allowance, protection/NTC details, connector and lead exit, MOQ and lead time. Include available test reports and transport documents; do not substitute a catalogue capacity for a fit-confirmed design.

The first meeting

We propose a 30-minute technical scoping call. Please identify the fastest way to establish compute/model feasibility, duplex call audio and mechanical/battery fit. If the planned early volumes are below your minimum, please state the commercial entry point or recommend a suitable prototype partner.

Source references

Manufacturer references (checked 10 September 2026):

lantronix.com/products/open-q-8550cs-som/

microchip.com/en-us/product/bm83

infineon.com/part/IM69D128S

adafruit.com/product/3923

ti.com/product/BQ51013C

tailscale.com/docs/features/tailscale-ssh

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